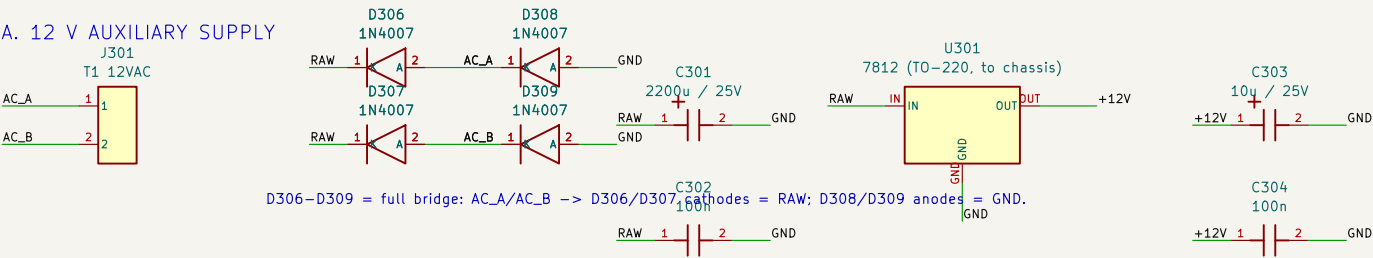


CTRL BOARD V0.6 / FIRST CIRCUIT DRAFT 2026-10-01 / UPC1237 protector + 12 V aux + soft-start timer / NO HARDWARE MEASUREMENT

Supply: T1 12 VAC winding (12.27 V no-load measured 2026-10-01, 1 A label) -> bridge -> 2200u -> 7812 -> +12V for relays, logic and UPC1237.

GND of this board ties to the PSU GND star at ONE point (J309). Relay coil current never flows through the amplifier signal ground (docs/13).

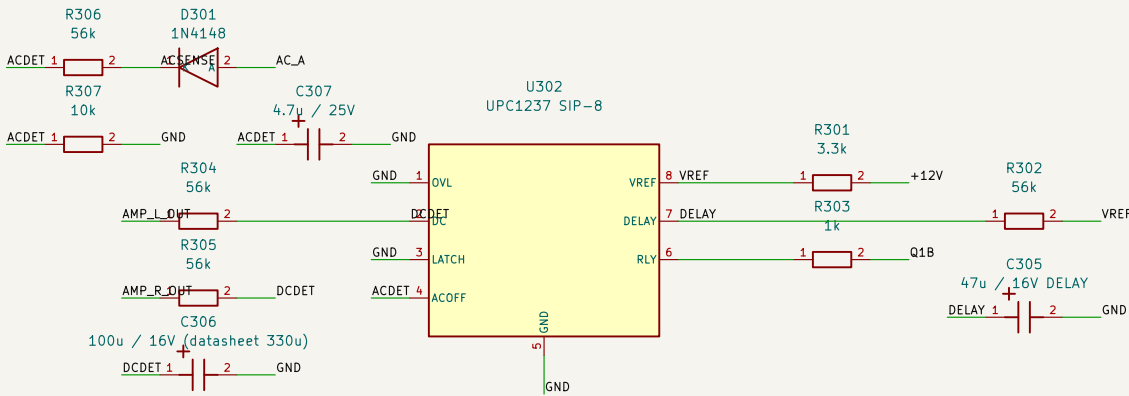
A. 12 V AUXILIARY SUPPLY



RAW -15-17 VDC at ~120 mA (3 x 33 mA G2RL coils + logic). 7812 dissipates ~0.6 W: clip heatsink or bolt to chassis. Ripple 2200u @120 mA ~0.5 Vpp (docs/14 6.5).

B. UPC1237 PROTECTOR (UTC QW-R107-065.B)

Thresholds: pin 2 +0.62 V / -0.17 V (DC), pin 4 0.74 V (AC-off), pin 8 = 3.4 V reference, pin 6 sinks <= 80 mA.

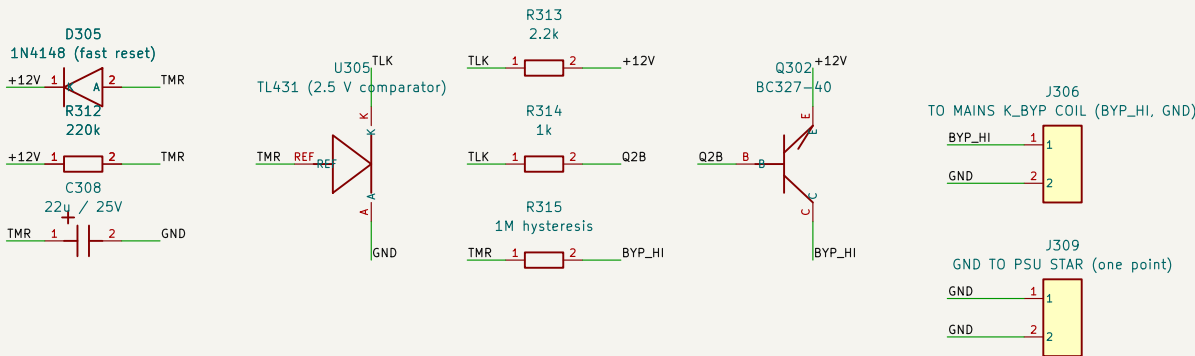


Pin 1 (overload) and pin 3 grounded: overload detector unused, automatic reset. For LATCH put 22n from pin 3 to GND instead.

DC trip (one channel, other at 0 V): +1.24 V / -0.34 V at the speaker output. 56k||56k x 100u = 2.8 s: 30 V fault trips in ~120 ms; 25 Vpk at 10 Hz leaves 0.07 V at pin 2 (<0.17 V).

AC-off: 17 Vpk x 10k/66k = 2.6 V at pin 4 while mains is on (droop between peaks to 1.8 V); after loss 10k x 4.7u decays below 0.74 V in ~60 ms (docs/14 6.4 budget <60 ms).

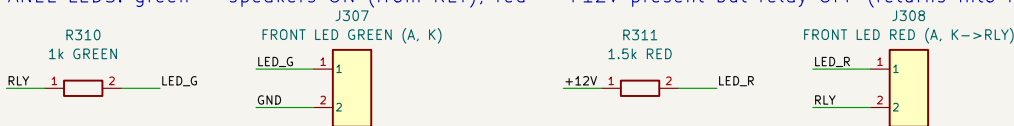
D. SOFT-START BYPASS TIMER: K_BYP (on MAINS board) pulls in ~1.1 s after +12V (docs/14 6.2 recommends 1 s)



t = 220k x 22u x ln(12/9.5) = 1.1 s. TL431 snaps at 2.5 V (sharp, no relay chatter); R315 adds ~0.3 V hysteresis; D305 empties C308 within ms at power-off so a quick off-on restarts the NTC limit.

Flyback diode for K_BYP sits on the MAINS board next to the coil (D403). K_TRIG is on the MAINS board, powered by the Z10 trigger, not by this board.

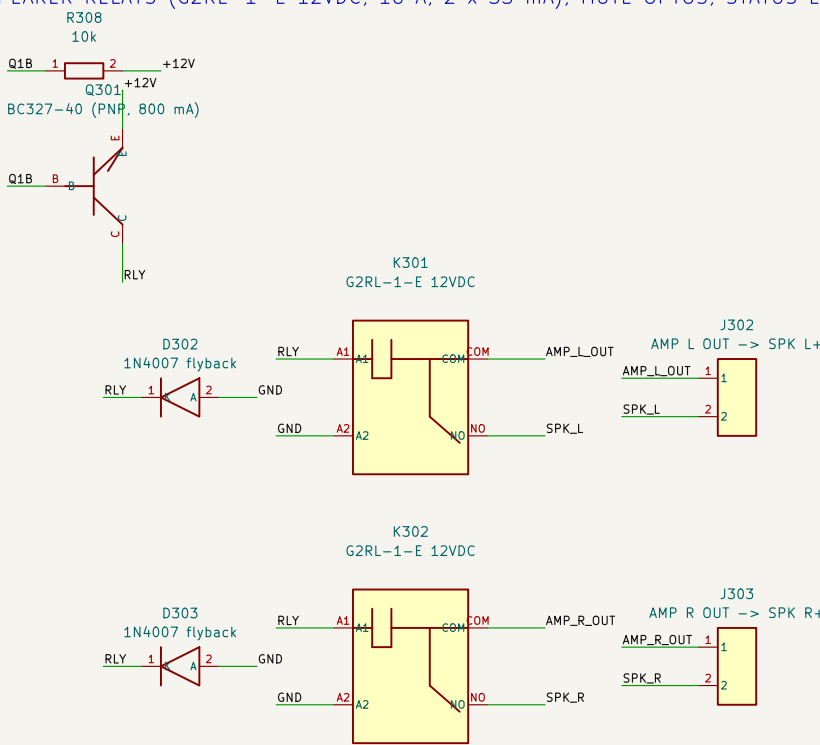
E. FRONT PANEL LEDS: green = speakers ON (from RLY), red = +12V present but relay OFF (returns into RLY)



Red LED returns into RLY: lit while RLY sits low (~6 mA through the coils, far below pull-in), dark once Q301 pulls RLY high.

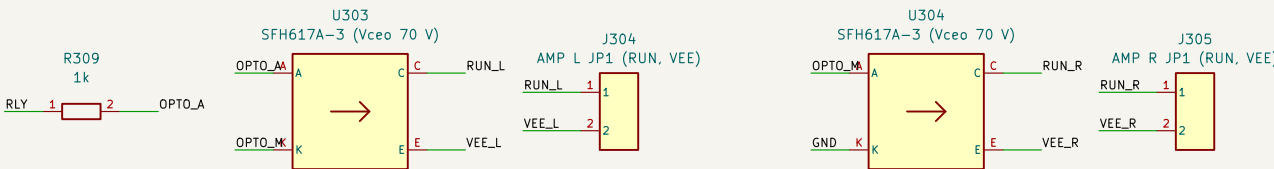
NOT DRAWN YET: single-rail-loss detector (docs/13 P5, optional), over-temperature (P6, optional). Component values are first-pass; every timing and threshold is to be measured per docs/14 9 before the board goes into the chassis.

C. SPEAKER RELAYS (G2RL-1-E 12VDC, 16 A, 2 x 33 mA), MUTE OPTOS, STATUS LEDS



Speaker return wire goes straight from the binding post to the amplifier J2.2 / J4.2, never through this board (docs/14 4).

Relay DC breaking: G2RL-1-E is rated 30 VDC; a 33.5 V / 8 ohm fault (4.5 A) is a one-shot protective break, accepted 2026-10-01 (docs/14 6.3 asks >=36 VDC).



Opto collector -> amplifier RUN (R8 end), emitter -> that board VEE: closes the JP1 position, 1.3 mA. Two independent paths; the L and R mute nodes stay isolated (docs/13).

Opto must stand |VEE| = 36 V when off: SFH617A-3 (70 V, DIP-4) or H11D1 (300 V, DIP-6); NOT PC817 (35 V). LED chain 2 x 1.2 V at ~8 mA from RLY through R309.

V0.6 control board / circuit draft only / no hardware measurements

Sheet: /
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Title: LM3886 V0.6 control / protection board (CTRL) – low voltage only

Size: A3 Date: 2026-10-01 Rev: 0.6 FIRST DRAFT
KiCad E.D.A. 10.0.6 Id: 1/1